

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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TASK 2 — HTML5 CANVAS AND CSS STYLING

1. TITLE

Implement HTML5 Canvas Graphics and Design a Styled Web Page Using CSS

2. AIM

To understand and implement **HTML5 Canvas** for drawing basic graphical elements and **CSS** for styling and designing a structured web page.

3. OBJECTIVE

The objectives of this task are:

- To understand the HTML5 <canvas> element.
 - To obtain the 2D drawing context using JavaScript.
 - To draw basic shapes using Canvas.
 - To draw a rectangle, circle, and line.
 - To display text using Canvas.
 - To understand external CSS styling.
 - To apply fonts, colors, spacing, borders, and shadows.
 - To create a navigation bar using CSS.
 - To design card-based content using Flexbox.
 - To apply hover effects to links and buttons.
 - To create a visually styled web page.
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4. THEORY

HTML5 Canvas

The **HTML5 Canvas** element provides a drawing area that can be controlled using JavaScript. It can be used to create graphics, shapes, text, animations, and other visual elements.

The <canvas> element defines the drawing area, while JavaScript is used to perform the actual drawing.

The `getContext("2d")` method obtains the **2D drawing context**, which provides methods for drawing shapes and text.

In this task:

- `fillRect()` is used to draw a rectangle.
- `arc()` is used to create a circular path.
- `stroke()` is used to draw the line.
- `fillText()` is used to display text.
- `fillStyle` is used to set the fill color.
- `font` is used to define the text font.

CSS

CSS (Cascading Style Sheets) is used to control the appearance and layout of HTML elements.

External CSS can be connected to an HTML page using the `<link>` element.

In this task, CSS is used for:

- Font styling
 - Background gradient
 - Text color
 - Navigation styling
 - Flexbox layout
 - Cards
 - Borders
 - Rounded corners
 - Box shadows
 - Button styling
 - Hover effects
 - Footer styling
-

5. ALGORITHM / PROCEDURE

Part A — Canvas

1. Create an HTML5 document.
2. Add a `<canvas>` element with a width of 500 pixels and height of 300 pixels.
3. Assign an ID to the canvas.
4. Use JavaScript to access the canvas element.
5. Obtain the 2D drawing context using `getContext("2d")`.
6. Set the fill color for the rectangle.
7. Draw a rectangle using `fillRect()`.
8. Create a circular path using `arc()`.
9. Fill the circle using `fill()`.
10. Create a line using `moveTo()` and `lineTo()`.
11. Draw the line using `stroke()`.
12. Set the font and text color.
13. Display text using `fillText()`.
14. Open the HTML file in a browser and verify the output.

Part B — CSS Styled Web Page

1. Create an HTML5 document.
2. Add a heading inside the header.

3. Create a navigation bar containing Home, About, Services, and Contact links.
 4. Create a container for the content cards.
 5. Add separate cards for HTML, CSS, and JavaScript.
 6. Add headings, descriptions, and buttons to each card.
 7. Create a footer.
 8. Create an external style.css file.
 9. Link the CSS file to the HTML document.
 10. Apply a gradient background to the page.
 11. Style the header and navigation links.
 12. Use Flexbox to arrange the cards.
 13. Apply borders, rounded corners, and shadows to the cards.
 14. Style the buttons.
 15. Add hover effects to navigation links and buttons.
 16. Open the HTML file in a browser.
 17. Verify the final styled webpage.
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6. PROGRAM

Part A — HTML5 Canvas

Task2_PastA.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Task2</title>
</head>
<body>
  <canvas id="myCanvas" width="500" height="300" style="border:1px solid black;">
</canvas>

<script>

var c=document.getElementById("myCanvas");

var ctx=c.getContext("2d");

//Rectangle

ctx.fillStyle="navy";

ctx.fillRect(20,20,120,80);

//Circle

ctx.beginPath();

ctx.arc(220,60,40,0,2*Math.PI);
```

```
ctx.fillStyle="Pink";

ctx.fill();

//Line

ctx.moveTo(20,150);

ctx.lineTo(250,150);

ctx.stroke();

//Text

ctx.font="25px Arial";

ctx.fillStyle="green";

ctx.fillText("Your Name",20,220);

</script>
</body>
</html>
```

Part B — HTML Page

Task2_partB.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My styled Webpg</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header>
<h1>My Styled Web Page</h1>
</header>
<nav>
<a href="#">Home</a>
<a href="#">About</a>
<a href="#">Services</a>
<a href="#">Contact</a>
```

```
</nav>
<div class="container">
<div class="card">
<h2>HTML</h2>
<p>HTML Structures Web Pages</p>
<button>Read More</button>
</div>
<div class="card">
<h2>CSS</h2>
<p>CSS Styles Web Page</p>
<button>Read More</button>
</div>
<div class="card">
<h2>JavaScript</h2>
<p>JavaScript adds interactivity</p>
<button>Read More</button>
</div>
</div>
<footer>
<p>Copyright 2026</p>
</footer>
</body> </html>
```

style.css

```
body{
  font-family: Arial, Helvetica, sans-serif;
  margin: 0;
  background: linear-gradient(to right, #996b84,#336674);
}
header{
  color: white;
  text-align: center;
  padding: 10px;
  text-shadow: 2px 2px 5px black;
}
nav{
  padding: 10px;
  text-align: center;
}
nav a{
  color: white;
  padding: 10px;
  text-decoration: none;
```

```
    text-shadow: 2px 2px 5px black;
}
nav a:hover{
    background: #743348;
    border-radius: 5px;
}
.container{
    display: flex;
    justify-content: space-around;
    margin: 20px;
}
.card{
    background: white;
    padding: 20px;
    border: 2px solid gray;
    border-radius: 15px;
    box-shadow: 5px 5px 10px gray;
    width: 250px;
    text-align: center;
}
button{
    padding: 10px;
    border: none;
    background: rgb(16, 106, 124);
    color: white;
}
button:hover{
    background: rgb(222, 23, 76);
    color: white;
    text-align: center;
    padding: 15px;
}
footer{
    color: white;
    text-align: center;
    padding: 15px;
    text-shadow: 2px 2px 5px black;
}
```

7. CODE EXPLANATION

Part A — Canvas

Code	Explanation
<canvas>	Creates a drawing area on the web page.
id="myCanvas"	Provides an identifier for accessing the canvas using JavaScript.
width="500"	Sets the canvas width to 500 pixels.
height="300"	Sets the canvas height to 300 pixels.
getElementById()	Retrieves the canvas element from the HTML document.
getContext("2d")	Obtains the 2D drawing context.
fillStyle	Specifies the fill color of graphical elements.
fillRect()	Draws and fills a rectangle.
beginPath()	Starts a new drawing path.
arc()	Creates an arc or circular path.
fill()	Fills the current path.
moveTo()	Sets the starting point of a line.
lineTo()	Specifies the ending point of a line.
stroke()	Draws the current path.
font	Specifies the font and size of text.
fillText()	Displays text on the canvas.

Part B — HTML and CSS

Code	Explanation
<link rel="stylesheet">	Connects the external CSS file to the HTML page.
<header>	Defines the header section.
<nav>	Defines the navigation section.
.container	Defines the main container for the cards.
display: flex	Creates a flexible layout for arranging elements.
justify-content: spacearound	Distributes the cards with space around them.
.card	Defines the appearance of individual content cards.
background	Sets the background color or gradient.
border	Adds a border around an element.
border-radius	Creates rounded corners.
box-shadow	Adds a shadow around an element.
text-shadow	Adds a shadow effect to text.
:hover	Applies styles when the mouse pointer moves over an element.
padding	Adds space inside an element.
margin	Adds space outside an element.

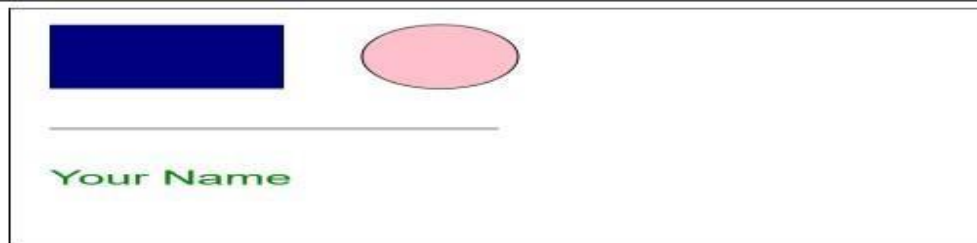
footer	Defines the footer section of the page.
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8. TEST CASES AND EXECUTION DATA

Part A — Canvas Test Cases

Test Case	TC-01
Test / Input	Load the webpage containing the graphical elements
Expected Result	The webpage should display a blue rectangle, pink circle, horizontal line, and "Your Name" text correctly.
Actual Result	All graphical elements and text were displayed correctly.
Status	Pass

TC-01 Output:



Part B — CSS Styled Web Page Output

Test Case	TC-02
Test / Input	Load the styled web page in the browser
Expected Result	The webpage should display the "My Styled Web Page" heading, navigation links, three content cards, and copyright text.
Actual Result	The complete styled webpage was displayed correctly.
Status	Pass

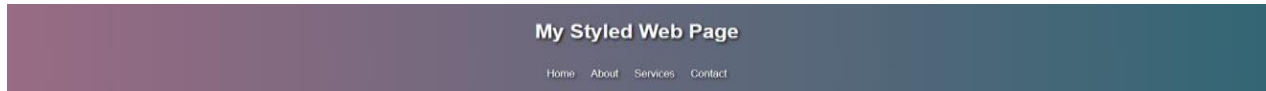
TC-02 Output:



Test Case	TC-03
Test / Input	Check the main heading and navigation bar

Expected Result	"My Styled Web Page" should be displayed with Home, About, Services, and Contact navigation links.
Actual Result	The heading and all navigation links were displayed correctly.
Status	Pass

TC-03 Output:



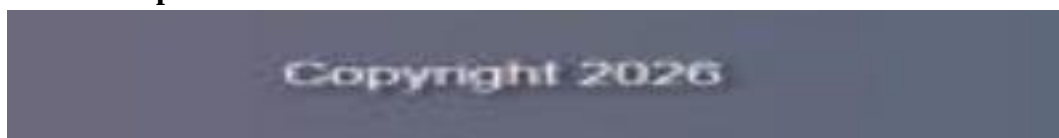
Test Case	TC-04
Test / Input	Check the HTML, CSS, and JavaScript content cards
Expected Result	Three cards labeled HTML, CSS, and JavaScript should be displayed with their respective descriptions.
Actual Result	All three cards and their descriptions were displayed correctly.
Status	Pass

TC-04 Output:



Test Case	TC-05
Test / Input	Check the copyright information
Expected Result	Copyright information should be displayed below the content cards.
Actual Result	"Copyright 2026" was displayed correctly.
Status	Pass

TC-05 Output:



9. OUTPUT

Part A — HTML5 Canvas Output

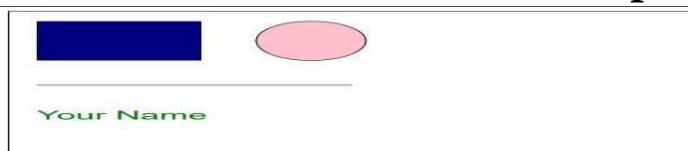


Figure 1: HTML5 Canvas output showing a rectangle, circle, line, and text.

Part B — CSS Styled Web Page Output



Figure 2: Styled web page showing header, navigation bar, HTML/CSS/JavaScript cards, buttons, and footer.

10. INTERVIEW QUESTIONS

1. Differentiate Margin and Padding.

Margin is the space outside an element, between the element and other elements.

Padding is the space inside an element, between the content and the border.

Code Snippet:

```
.container {  
    margin: 20px;  
}  
  
.card {  
    padding: 20px;  
}
```

In the given program, `margin: 20px` is used for the container, while `padding: 20px` is used inside each card.

2. Explain the difference between Border and Outline.

A Border is the boundary around an element and is part of the CSS box model.

An Outline is drawn outside the border and does not occupy space or affect the layout of the element.

Code Snippet:

```
.card {  
    border: 2px solid gray;  
    border-radius: 15px;  
}
```

In the given program, a border is used around each card to define its boundary. No outline is used in this program.

3. What is the purpose of the Canvas element?

The HTML5 `<canvas>` element provides a drawing area that can be used with JavaScript to create graphics such as rectangles, circles, lines, and text.

Code Snippet:

```
<canvas id="myCanvas" width="500" height="300"
      style="border:1px solid black;">
</canvas>
```

JavaScript:

```
var c = document.getElementById("myCanvas");
var ctx = c.getContext("2d");

ctx.fillStyle = "navy";
ctx.fillRect(20,20,120,80);

ctx.beginPath();
ctx.arc(220,60,40,0,2*Math.PI);
ctx.fillStyle = "Pink";
ctx.fill();

ctx.moveTo(20,150);
ctx.lineTo(250,150);
ctx.stroke();

ctx.font = "25px Arial";
ctx.fillStyle = "green";
ctx.fillText("Your Name",20,220);
```

In the given program, Canvas is used to draw a rectangle, circle, line, and text.

4. Compare Inline, Internal, and External CSS.

Inline CSS is written directly inside an HTML element using the `style` attribute.

```
<p style="color: red;">Hello</p>
```

Internal CSS is written inside the `<style>` element in the HTML document.

```
<style>
  p {
    color: red;
  }
</style>
```

External CSS is written in a separate `.css` file and linked to the HTML document.

```
<link rel="stylesheet" href="style.css">
```

In the given program, **External CSS** is used because all the styling is written separately in `style.css`, making the HTML code cleaner and the styles reusable.

5. How do Hover effects improve user experience?

Hover effects provide visual feedback when the user moves the pointer over an interactive element. They help users identify clickable buttons and navigation links and make the webpage more interactive.

Code Snippet:

```
nav a:hover {  
    background: #743348;  
    border-radius: 5px;  
}  
  
button:hover {  
    background: rgb(222, 23, 76);  
    color: white;  
    text-align: center;  
    padding: 15px;  
}
```

In the given program, hover effects are applied to navigation links and buttons to make them visually different when the user points at them.

11. RESULT

Thus, **HTML5 Canvas graphics** were successfully implemented using JavaScript to draw a rectangle, circle, line, and text. A **styled web page** was also successfully developed using external CSS, including Flexbox layout, navigation styling, cards, borders, shadows, gradients, buttons, and hover effects. The outputs were verified successfully.